StudySapiens Solved Notes

Ch.1 Biodiversity & Classification

Biology 11 • Punjab English Medium • PECTAA 2025–26 • Solved Exercise
Prompts are paraphrased and keyed to textbook question numbers. MCQ answer letters refer to the textbook options.

SECTION 1:MULTIPLE CHOICE QUESTIONS

Answer letters refer to the textbook options. Full textbook stems/options are not reproduced here.

1. Which domain contains many organisms of extreme habitats and has cell membranes with ether-linked lipids?

Answer: (b) Archaea — Archaea have ether-linked membrane lipids and include many extremophiles; Bacteria have ester-linked lipids and peptidoglycan walls.

2. Which feature correctly separates the domains Bacteria and Archaea?

Answer: (b) Peptidoglycan in bacterial cell walls — Bacterial cell walls contain peptidoglycan, while archaeal cell walls lack peptidoglycan.

3. Which kingdom includes mostly unicellular eukaryotes that may be autotrophic or heterotrophic?

Answer: (d) Protoctista — Protista/Protoctista includes mostly unicellular eukaryotes such as protozoa and algae; some are autotrophic and others heterotrophic.

4. Which kingdom is multicellular, autotrophic and has cell walls made of cellulose?

Answer: (c) Plantae — Plants are multicellular eukaryotes with cellulose cell walls and prepare food by photosynthesis.

5. Which criterion is commonly used for classifying viruses?

Answer: (b) The type of nucleic acid they contain — Viruses are classified by genetic material (DNA or RNA), replication strategy, morphology and host range; particle colour is not a classification criterion.

6. Coronaviruses and influenza viruses belong to which virus group?

Answer: (d) Single-stranded RNA viruses — Both coronaviruses and influenza viruses have single-stranded RNA genomes.

7. Which level of biodiversity describes the variety of species in a habitat or ecosystem?

Answer: (c) Species diversity — Species diversity includes the number of species and their relative abundance/distribution in an area.

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8. Which method is most suitable for studying species distribution along an environmental gradient?

Answer: (c) Transect sampling — A transect records organisms at intervals along a line or strip, so it shows changes across a gradient.

9. Which statement agrees with the biological species concept?

Answer: (c) Members of the same species are reproductively isolated from other species — The biological species concept defines species as interbreeding natural populations that are reproductively isolated from other such groups.

10. Which type of speciation occurs when a physical barrier geographically separates populations?

Answer: (c) Allopatric speciation — Geographic isolation stops gene flow; separated populations diverge until reproductive isolation develops.

SECTION 2: SHORT QUESTIONS

1. Name the three domains of life. How do they differ in cellular structure?

The three domains are **Bacteria**, **Archaea** and **Eukarya**. Bacteria and Archaea are prokaryotic, so they lack a true nucleus and membrane-bound organelles. Bacteria usually have peptidoglycan in their cell walls and ester-linked membrane lipids. Archaea lack peptidoglycan and have ether-linked lipids; many live in extreme habitats. Eukarya have a true nucleus and membrane-bound organelles, and may be unicellular or multicellular.

2. Give one key feature that separates Archaea from Bacteria.

A key difference is the cell wall. Bacterial cell walls contain **peptidoglycan**, whereas archaeal cell walls do not contain peptidoglycan. Archaea may have other wall materials, and their membrane lipids are ether-linked rather than ester-linked.

3. How do sponges reproduce asexually?

Sponges reproduce asexually mainly by **budding** and **regeneration**. Some sponges also form resistant capsules called **gemmules**, which contain masses of amoeboid cells. When conditions become favourable, the cells come out of the gemmule and develop into a new sponge.

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4. Differentiate the following.

(i) Sac-like and tube-like digestive system

Feature	Sac-like digestive system	Tube-like digestive system
Openings	One opening serves as both mouth and anus.	Two openings: mouth and anus.
Food passage	Food enters and undigested waste leaves through the same opening.	Food moves in one direction along a complete alimentary canal.
Efficiency	Less efficient; mixing of incoming food and waste is possible.	More efficient; allows regional specialization of digestion and absorption.
Examples	Cnidarians such as Hydra; flatworms such as Planaria.	Most higher animals, e.g., annelids, arthropods and chordates.

(ii) Polyps and medusae

Feature	Polyp	Medusa
Body form	Cylindrical, usually sessile body attached to a surface.	Umbrella- or bell-shaped, usually free-swimming body.
Mouth/tentacles	Mouth and tentacles usually directed upward.	Mouth usually on the lower side; tentacles around the bell margin.
Reproduction	Commonly reproduces asexually by budding.	Commonly reproduces sexually by gametes.
Examples	Hydra, sea anemone, corals.	Jellyfish.

(iii) Urochordata and cephalochordate

Feature	Urochordata	Cephalochordata
Notochord	Present only in the tail of the free-swimming larva; usually lost in the adult.	Extends from head to tail and persists throughout life.
Adult body	Adult is usually sessile and covered by a tunic.	Adult is fish-like and retains chordate features.
Examples	Sea squirts (tunicates).	Amphioxus (Branchiostoma).

(iv) Ectotherms and endotherms

Feature	Ectotherms	Endotherms
Heat source	Mainly depend on external environmental heat.	Produce most body heat internally by metabolism.
Body temperature	Body temperature varies with the environment.	Maintain a relatively constant core temperature.
Energy need	Generally need less food/energy.	Generally need more food/energy to maintain temperature.
Examples	Fishes, amphibians, reptiles.	Birds and mammals.

(v) Marsupial and placental mammals

Feature	Marsupials	Placental mammals
Young at birth	Give birth to very immature young; development continues in a pouch (marsupium).	Young develop longer inside the uterus and are born more developed.
Placenta	Placenta is short-lived/simple or less developed.	Well-developed placenta nourishes the foetus and removes wastes.
Examples	Kangaroo, opossum, koala.	Human, dog, whale, bat, elephant.

(vi) Gymnosperms and angiosperms

Feature	Gymnosperms	Angiosperms
Seeds	Seeds are naked, usually borne on cones.	Seeds are enclosed within an ovary/fruit.
Flowers/fruits	Do not produce true flowers or fruits.	Produce flowers and fruits.
Examples	Pine, spruce, cycads, Ginkgo.	Grasses, corn, rose, sunflower, mango.

5. Which kingdom is characterized by chitin cell walls and mostly decomposers?

Kingdom Fungi. Fungi are eukaryotic heterotrophs with cell walls made of chitin. They absorb nutrients from their surroundings and many are saprophytic decomposers, although some are parasites or mutualists.

6. What type of speciation occurs when populations are geographically separated?

Allopatric speciation. A geographic barrier separates populations and prevents gene flow. Over time, mutation, natural selection and genetic drift make the populations different enough that they can no longer interbreed successfully.

7. What is the role of genetic drift in speciation?

Genetic drift is a random change in allele frequencies, especially in small populations. It can cause isolated populations to lose or fix alleles by chance, reducing genetic variation and increasing divergence. Over time, this can contribute to reproductive isolation and speciation, particularly after founder events or bottlenecks.

8. Which primary method is used to study species distribution along an environmental gradient?

Transect sampling. A line or strip is laid across the study area and species are recorded at regular intervals. This shows how species composition and abundance change along gradients such as shore height, altitude or soil moisture.

9. Which level of biodiversity assessment involves the variety of ecosystems in a region?

Ecosystem diversity. It assesses the range of habitats and ecosystems, such as forests, wetlands, grasslands and deserts, and how they function and contribute to ecological health.

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SECTION 3: LONG QUESTIONS

1. Compare and contrast Archaea and Bacteria. What does this tell us about their evolutionary histories?

Archaea and Bacteria are both prokaryotic domains, but they differ in important structural and molecular features. These differences show that they are separate evolutionary lineages.

Feature	Bacteria	Archaea
Cell type	Prokaryotic; no true nucleus or membrane-bound organelles.	Prokaryotic; no true nucleus or membrane-bound organelles.
Cell wall	Usually contains peptidoglycan.	Lacks peptidoglycan; may contain other polysaccharides/proteins.
Membrane lipids	Ester-linked, unbranched fatty acid lipids.	Ether-linked, branched hydrocarbon lipids; more stable in extreme conditions.

Feature	Bacteria	Archaea
Genetic machinery	Simpler RNA polymerase; generally no histones.	More complex RNA polymerase; some have histone-like proteins; shares some features with eukaryotes.
Habitats	Widely distributed in soil, water and living hosts.	Many live in extreme habitats, but also occur in ordinary environments.
Reproduction	Mainly binary fission.	Mainly binary fission or similar asexual division.

Evolutionary meaning: the three-domain system separates Bacteria from Archaea because their molecular differences are deep. Archaea share some information-processing features with Eukarya, suggesting that Archaea are more closely related to eukaryotes than to Bacteria. Thus, prokaryotes are not one uniform group; they represent two ancient domains with different evolutionary histories.

2. Explain the biological species concept. How does it define species boundaries and speciation? Give examples.

The **biological species concept** defines a species as a group of natural populations whose members can interbreed and produce fertile offspring, and which are reproductively isolated from other such groups. Reproductive isolation is the key boundary: members of different species do not normally mate, or their offspring are not viable/fertile.

Speciation occurs when populations become reproductively isolated and accumulate genetic differences. Geographic isolation can start allopatric speciation; isolation can also arise within the same area through polyploidy, niche differentiation or behavioural changes.

- **Example of species boundary:** a horse and a donkey can mate, but their offspring (mule) is usually sterile, so they are treated as different species.
- **Example of speciation:** Darwin's finches on the Galápagos Islands diverged after populations became isolated on different islands and adapted to different foods/environments.

Limitation: the concept is difficult to apply to asexual organisms, fossils and cases where hybridization occurs.

3. Describe the major groups of Kingdom Protista and Kingdom Fungi.

Kingdom Protista

Protista includes eukaryotes that are unicellular, colonial, filamentous or simple multicellular. The major groups are:

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Group	Main features	Examples
Protozoa	Animal-like, unicellular, heterotrophic protists.	Amoeba, Paramecium, Plasmodium, Trypanosoma.
Algae	Plant-like protists with chlorophyll; autotrophic; many have cellulose cell walls.	Euglena, diatoms.
Fungus-like protists (Myxomycota and Oomycota)	Hyphae-like structures; mostly saprophytic.	Slime molds and water molds.

Kingdom Fungi

Fungi are eukaryotic heterotrophs with chitin cell walls. They absorb food from the surroundings. The major groups are:

Group	Main features	Examples
Zygomycota	Hyphae usually lack septa (coenocytic).	Rhizopus (bread mold).
Ascomycota	Largest fungal group; septate hyphae.	Yeasts, morels, truffles, cup fungi, Neurospora, common molds.
Basidiomycota	Septate hyphae.	Mushrooms, toadstools, puffballs, bracket fungi, rusts and smuts.

4. Describe the general characteristics of the following phyla.

(i) Phylum Cnidaria

- Mostly marine; radial symmetry; diploblastic (epidermis and gastrodermis with mesoglea between).
- Possess cnidocytes containing nematocysts for defence and prey capture.
- Have a sac-like gastrovascular cavity with a single opening (mouth); no anus.
- Body forms include polyp and medusa; nerve net present; no complex organ systems.
- Examples: Hydra, jellyfish, sea anemone, corals, Obelia.

(ii) Phylum Platyhelminthes

- Bilateral symmetry; triploblastic; acoelomate; dorsoventrally flattened body.
- Organ-system level of organization; incomplete digestive tract or absent in tapeworms.
- Excretion/osmoregulation by protonephridia with flame cells.
- Many are parasitic; free-living forms such as Planaria occur in water.
- Examples: Planaria, liver fluke, tapeworm.

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(iii) Phylum Mollusca

- Bilateral symmetry; triploblastic; coelomate; soft body usually divided into head, muscular foot and visceral mass.
- A mantle covers the visceral mass and often secretes a calcareous shell.
- Most have a radula for feeding (absent in bivalves); circulation is usually open.
- Respiration by gills or lung-like structures; many have well-developed sense organs.
- Examples: snails, clams, octopus, squid.

(iv) Phylum Arthropoda

- Bilateral symmetry; triploblastic; coelomate; segmented body with jointed appendages.
- Chitinous exoskeleton; growth occurs by moulting (ecdysis).
- Open circulatory system; respiration by gills, tracheae or book lungs depending on group.
- Well-developed sensory organs; great diversity and ecological success.
- Examples: insects, spiders, scorpions, crabs, prawns, centipedes.

(v) Phylum Echinodermata

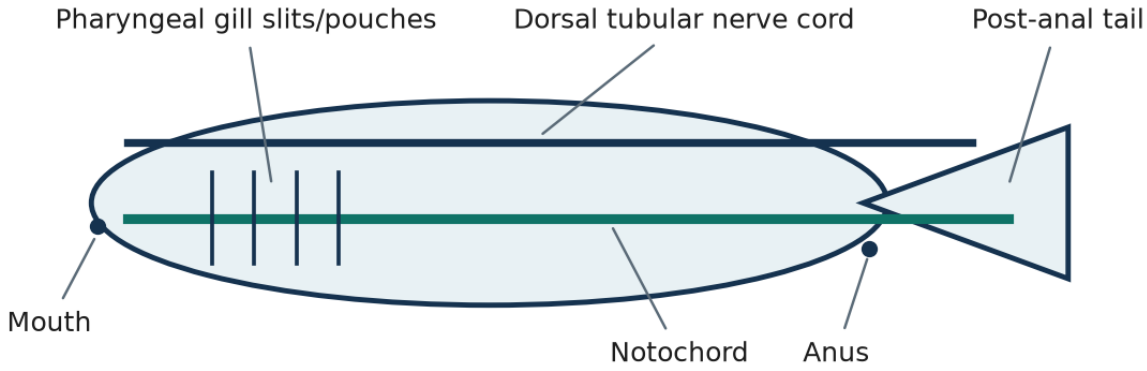
- Exclusively marine; adults usually pentaradial symmetry; larvae bilateral.
- Triploblastic, coelomate; endoskeleton of calcareous ossicles/spines.
- Water vascular system with tube feet for locomotion, feeding and gas exchange.
- No distinct head; good powers of regeneration in many forms.
- Examples: starfish, sea urchins, sea cucumbers, brittle stars.

5. Describe the general characteristics of Phylum Chordata.

Chordates are bilateral, triploblastic, coelomate animals. At least at some stage of life, they show the following diagnostic features:

- **Notochord:** a flexible supporting rod along the dorsal side; in vertebrates it is largely replaced by the vertebral column.
- **Dorsal tubular nerve cord:** lies above the notochord; its anterior end forms the brain in vertebrates.
- **Pharyngeal gill slits/pouches:** appear in the pharynx; they may develop into gills or be modified in land vertebrates.
- **Post-anal tail:** extends beyond the anus; it may persist or degenerate during development.

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Simplified chordate body plan showing the diagnostic features.

Other common features include a closed circulatory system, ventral heart, segmented muscles and an internal skeleton. The phylum includes invertebrate chordates and vertebrates.

Group	Key point	Examples
Subphylum Urochordata	Notochord and nerve cord mainly in the larval tail; adults usually sessile tunicates.	Sea squirts.
Subphylum Cephalochordata	Notochord persists throughout life and extends into the head region.	Amphioxus.
Vertebrates	Notochord replaced by vertebral column; cranium present.	Fishes, amphibians, reptiles, birds, mammals.

6. Describe the three groups of mammals.

Mammals are endothermic vertebrates with hair and mammary glands. They have a four-chambered heart, a muscular diaphragm, and usually give birth to young (except egg-laying mammals). The three groups are:

Group	Main characteristics	Examples
Monotremes (Prototheria)	Egg-laying mammals; oviparous; young develop inside eggs outside the mother’s body.	Duckbill platypus, echidna (spiny anteater).
Marsupials (Metatheria)	Give birth to immature young; development continues in a pouch (marsupium); placenta simple/short-lived.	Kangaroo, opossum, koala.

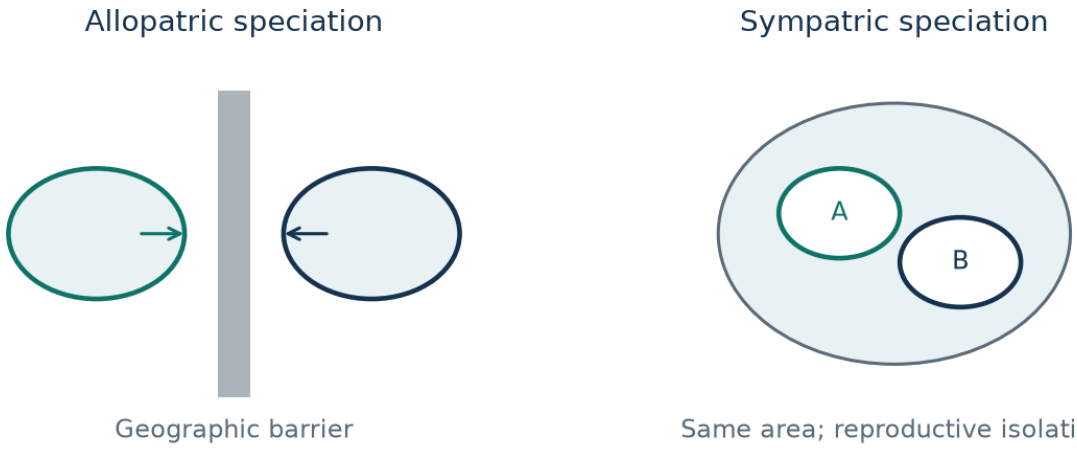
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Group	Main characteristics	Examples
Placental mammals (Eutheria)	Well-developed placenta connects foetus with uterine wall for nutrition, gas exchange and waste removal; young are born more developed.	Human, rat, monkey, bat, elephant, dolphin.

All three groups share mammalian features such as mammary glands, hair, skin glands and three middle-ear bones, but they differ mainly in reproduction and early development.

7. Explain the mechanisms of allopatric and sympatric speciation.

Feature	Allopatric speciation	Sympatric speciation
Starting condition	A population is divided by a geographic barrier.	New species arise within the same geographic area.
Gene flow	Gene flow is stopped or greatly reduced by the barrier.	Gene flow is reduced by reproductive barriers, not by physical separation.
Main mechanisms	Mutation, natural selection and genetic drift act independently in isolated populations.	Polyploidy, niche differentiation, behavioural/temporal isolation or disruptive selection.
Result	Populations diverge until they cannot interbreed even if the barrier is removed.	Reproductive isolation develops while populations remain in the same area.
Examples	Darwin's finches on different Galápagos islands.	Polyploid plants; insects shifting to different host plants/niches.



Allopatric speciation involves geographic separation; sympatric speciation occurs without a physical barrier.

In both cases, the essential outcome is **reproductive isolation**. Once gene flow is sufficiently reduced, genetic differences can accumulate until new species are

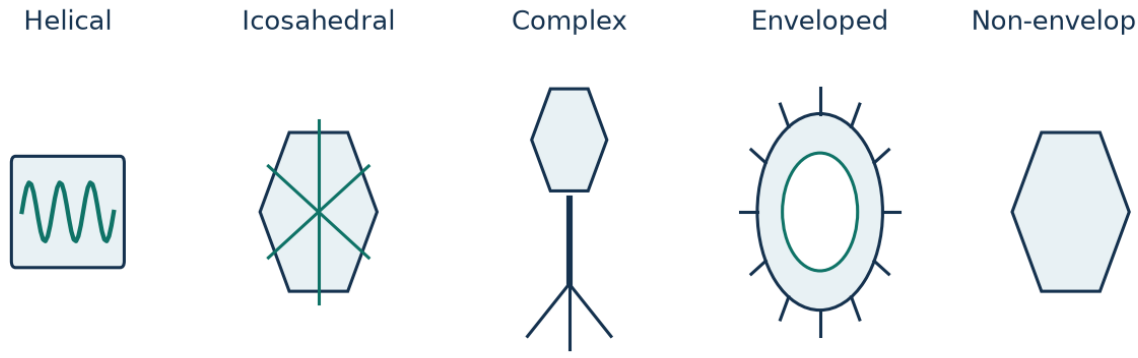
formed.

8. How are viruses classified on the basis of structure and replication strategy? What is the significance of this classification in virology?

Viruses are acellular and are classified by several features, including genetic material, morphology, presence of an envelope, host range and replication strategy.

Classification by structure

Basis	Groups	Examples
Morphology	Helical, icosahedral, complex.	Tobacco mosaic virus (helical); adenovirus (icosahedral); bacteriophage (complex).
Envelope	Enveloped and non-enveloped (naked).	Influenza virus/HIV (enveloped); poliovirus/adenovirus (naked).
Genetic material	DNA or RNA; single-stranded or double-stranded.	Adenovirus/herpesvirus (dsDNA); parvovirus (ssDNA); coronavirus/influenza (ssRNA); rotavirus (dsRNA).



Common structural forms of viruses.

Classification by replication strategy

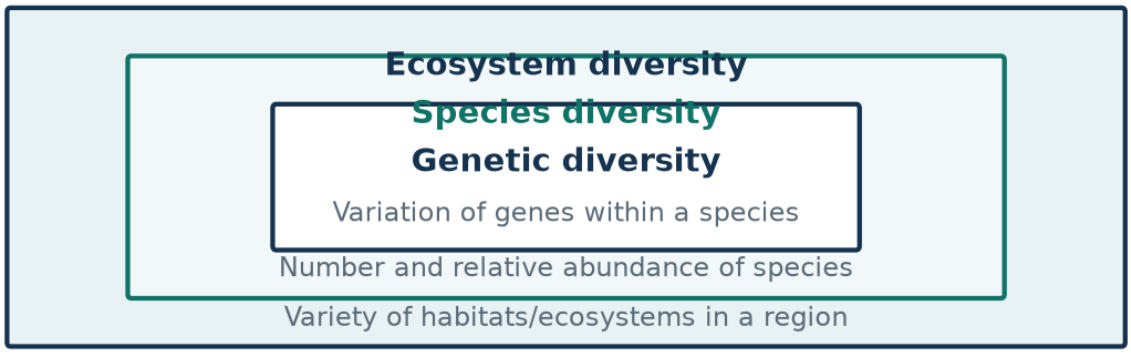
- **Positive-sense RNA viruses:** their RNA can be directly translated by host ribosomes; e.g., poliovirus, hepatitis C virus.
- **Negative-sense RNA viruses:** their RNA must first be transcribed into mRNA by a viral RNA polymerase; e.g., rabies virus, Ebola virus.
- **Reverse-transcribing viruses:** replicate through a DNA intermediate using reverse transcriptase; e.g., HIV (RNA genome) and hepatitis B virus (DNA genome).

Significance: this classification helps virologists understand how viruses replicate, evolve and cause disease. It also guides diagnosis, antiviral drug targets, vaccine development and prediction of host range and transmission.

9. What are the levels of biodiversity assessment? How do they help us understand diversity and conservation?

Biodiversity is assessed at three main levels:

- **Genetic diversity:** variation of genes within a species. It helps populations adapt to environmental change and resist disease.
- **Species diversity:** the number of species and their relative abundance/distribution in an area. It shows community richness and evenness.
- **Ecosystem diversity:** the variety of habitats/ecosystems in a region, such as forests, wetlands, grasslands and deserts. It shows how different systems function and support life.



Biodiversity can be assessed from genes to species to ecosystems.

Conservation importance: assessing all three levels helps identify areas of high conservation value, protect endangered species and genetic variation, monitor ecosystem health, and maintain services such as pollination, nutrient cycling and climate regulation.

10. Why are random sampling methods important in ecological studies?

Random sampling gives every part of a study area an equal chance of being sampled. This reduces bias and makes the sample more representative of the whole community.

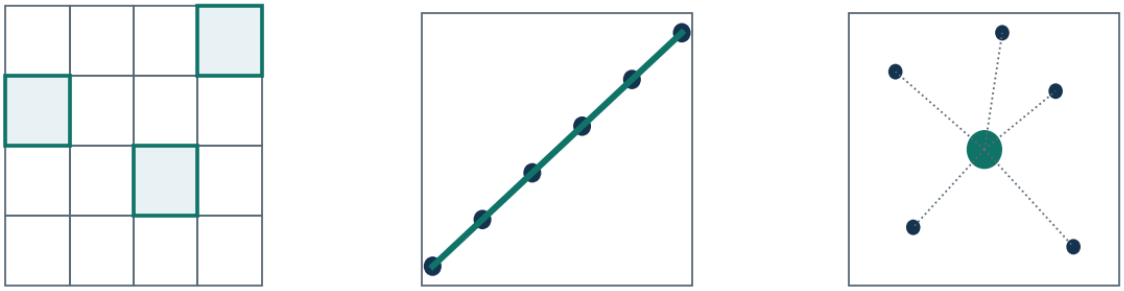
- It minimizes selection bias and gives a more accurate picture of biodiversity.
- It provides reliable data for statistical estimates of species richness, abundance and distribution.
- It allows standardized comparisons between different habitats or areas.
- It improves representativeness by covering different parts of the study area.
- It supports conservation by identifying valuable areas and monitoring change over time.

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Common methods

Method	Use
Quadrat sampling	Sampling randomly selected squares; useful for plants and sessile organisms.
Transect sampling	Recording species along a line/strip; useful across environmental gradients.
Aerial surveys / remote sensing	Observing large or inaccessible areas using aircraft, drones or satellite imagery.
Point counts	Counting organisms seen/heard from a fixed point; common for birds.
Mark–recapture	Estimating population size of mobile animals by marking, releasing and recapturing.

Quadrat sampling Transect sampling Point count / aerial survey



Examples of field methods used to assess distribution and abundance.

11. Explain the concept of ecosystem and niche.

An **ecosystem** is a dynamic, interactive system made up of living organisms (biotic factors) and their physical environment (abiotic factors). Energy flows and nutrients cycle through the interactions between organisms and their environment. Examples include a pond, forest or desert.

A **niche** is the role or function of a species within an ecosystem. It includes the habitat it uses, the resources it needs, its interactions with other organisms (predation, competition, symbiosis) and its role in energy flow.

Concept	Meaning	Example
Ecosystem	Community of organisms plus abiotic environment, interacting as a system.	A pond with algae, insects, fish, water, light and minerals.
Niche	The functional role and resource use of a species in that system.	A fish species feeding on insects, living in open water and serving as prey for birds.

Understanding ecosystems and niches helps explain species coexistence, competition and the stability of ecological communities.

INQUISITIVE QUESTIONS

1. How does classification of viruses by nucleic acid content and replication method help us understand viral diversity?

Viruses may have DNA or RNA, and their genomes may be single-stranded or double-stranded. Their replication strategies also differ: some RNA genomes are translated directly, some must first be copied into mRNA, and reverse-transcribing viruses use a DNA intermediate. These differences reveal major evolutionary and functional groups of viruses. They also explain differences in mutation rate, host interaction, disease patterns and possible treatments, so classification helps organize viral diversity in a meaningful way.

2. What is a drawback of the biological species concept?

The main drawback is that it cannot be applied easily to organisms that do not reproduce sexually, such as many bacteria and other asexual organisms. It is also difficult to use for fossils/extinct organisms and for cases where distinct species hybridize or where reproductive isolation is hard to observe. Therefore, species boundaries sometimes require additional evidence such as morphology, genetics or ecology.

3. How does biodiversity contribute to ecosystem balance?

Biodiversity supports ecosystem balance by maintaining food webs, nutrient cycling, pollination, seed dispersal and natural pest control. Genetic diversity helps populations adapt to change, species diversity provides functional redundancy and resilience, and ecosystem diversity maintains different habitats and services. When biodiversity is lost, ecosystems become less stable and more vulnerable to disturbance, disease and collapse.

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